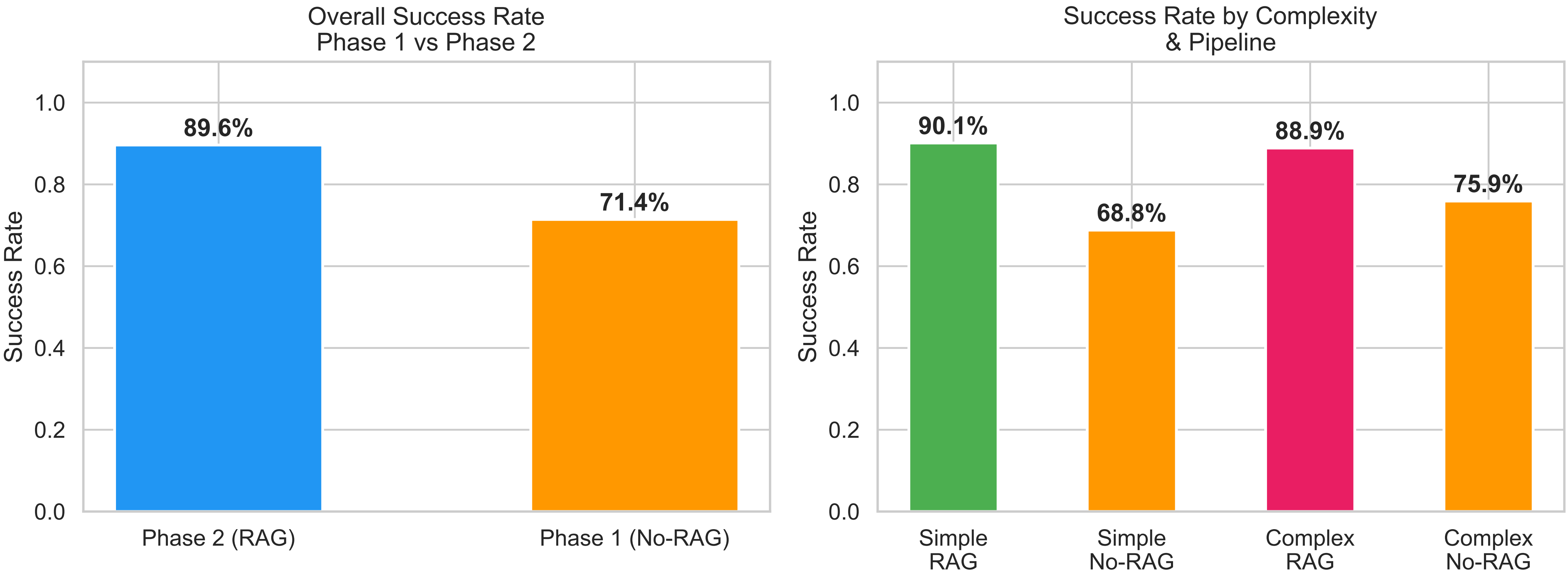
Figure 1 — Generation Success Rate Comparison



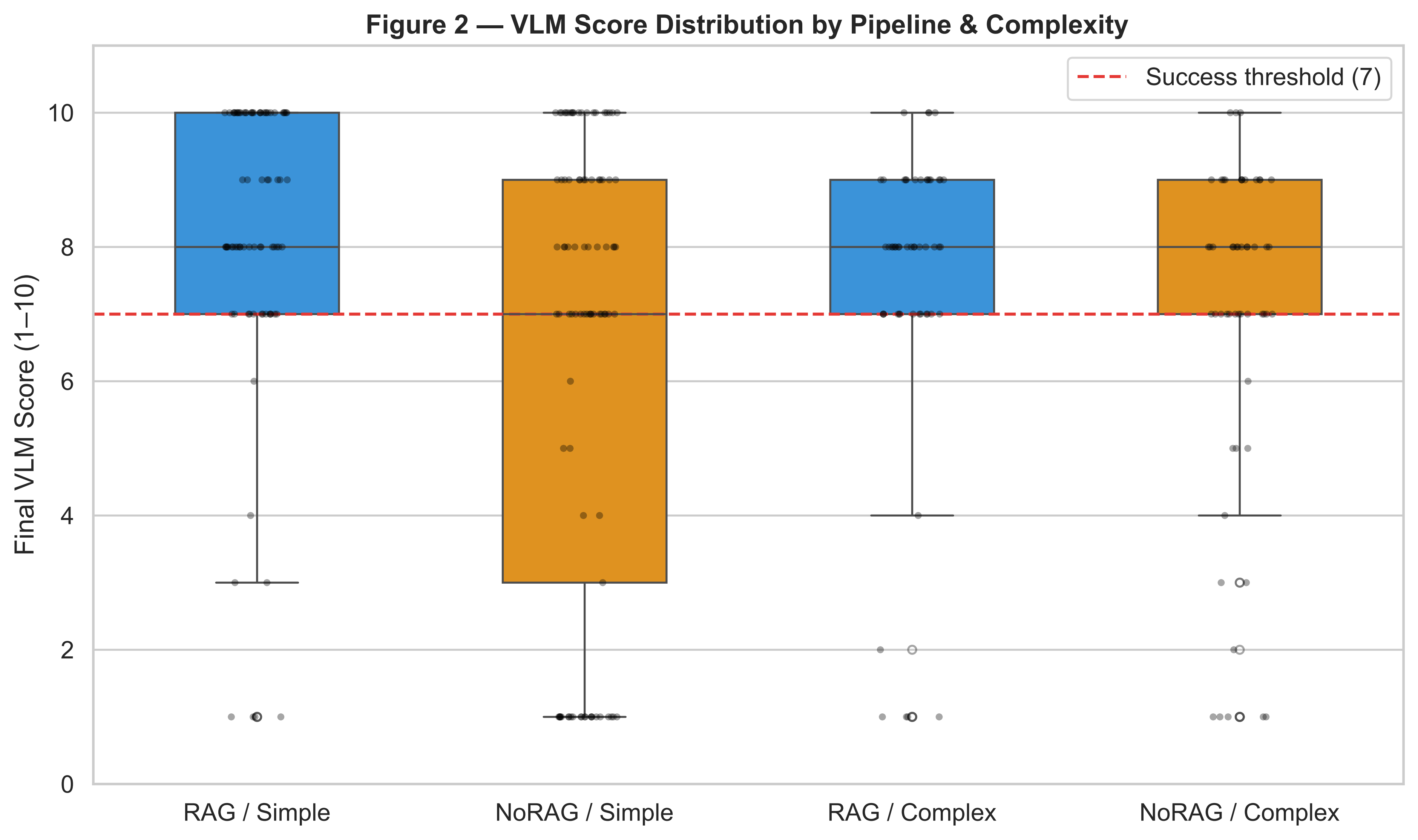


Figure 2 — VLM Score Distribution by Pipeline & Complexity
Success threshold (7)
Final VLM Score (1–10)
0
2
4
6
8
10
RAG / Simple
NoRAG / Simple
RAG / Complex
NoRAG / Complex

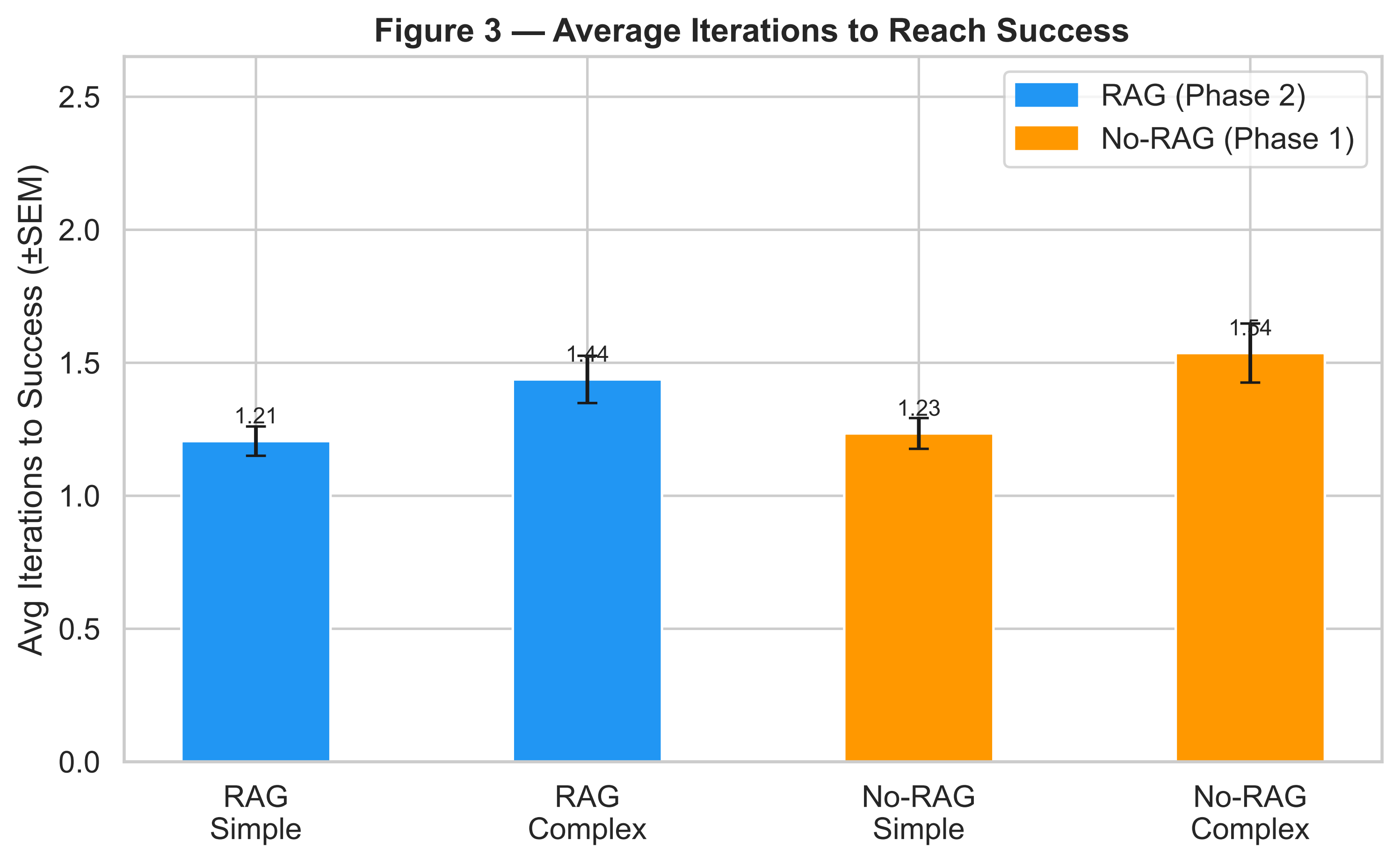
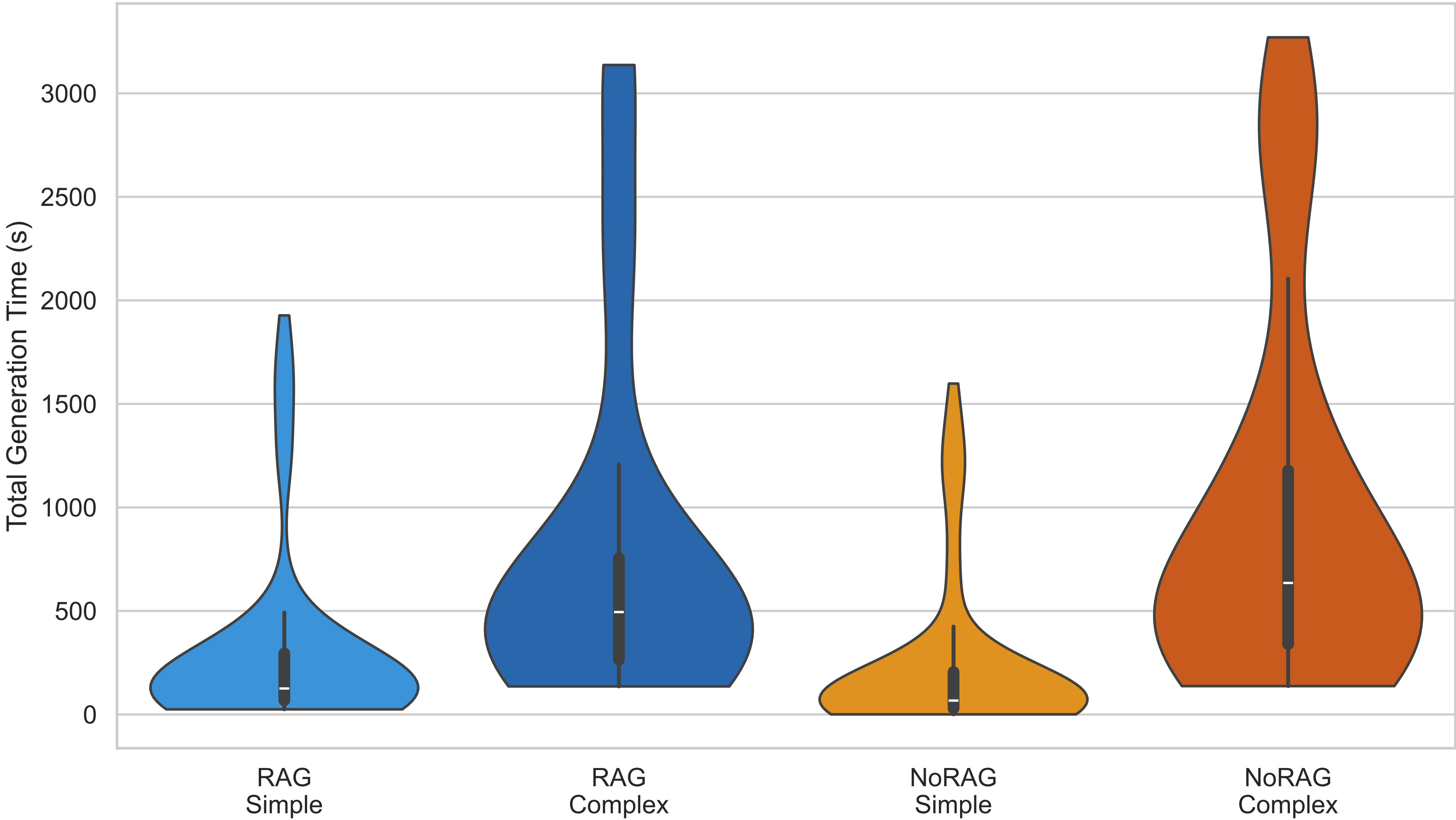


Figure 3 — Average Iterations to Reach Success
Avg Iterations to Success (±SEM)
0.0
0.5
1.0
1.5
2.0
2.5
RAG (Phase 2)
No-RAG (Phase 1)
1.21
1.44
1.23
1.54
RAG
Simple
RAG
Complex
No-RAG
Simple
No-RAG
Complex

Figure 4 — Generation Time Distribution (Violin)



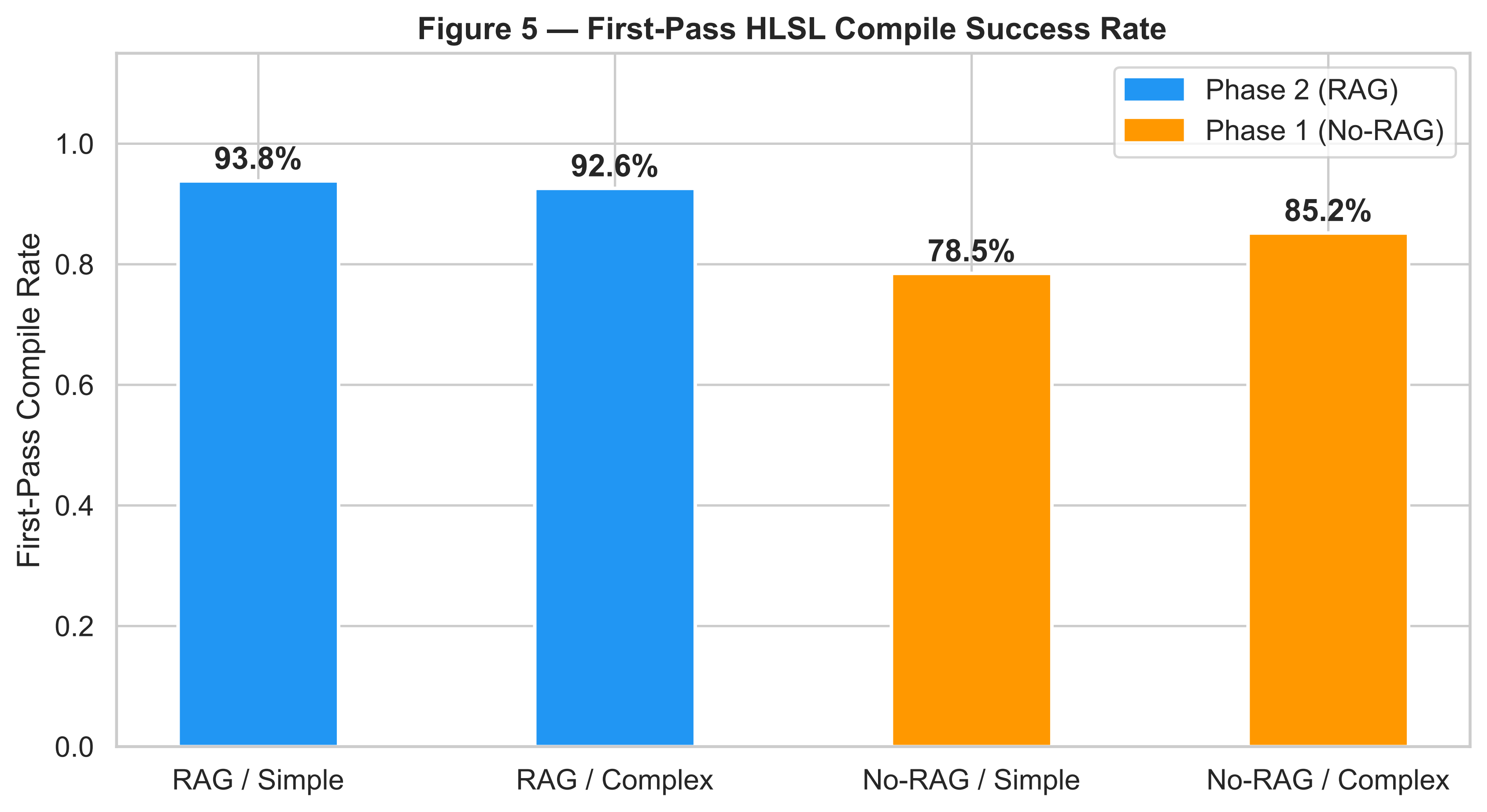


Figure 5 — First-Pass HLSL Compile Success Rate
First-Pass Compile Rate
1.0
0.8
0.6
0.4
0.2
0.0
93.8%
92.6%
78.5%
85.2%
Phase 2 (RAG)
Phase 1 (No-RAG)
RAG / Simple
RAG / Complex
No-RAG / Simple
No-RAG / Complex

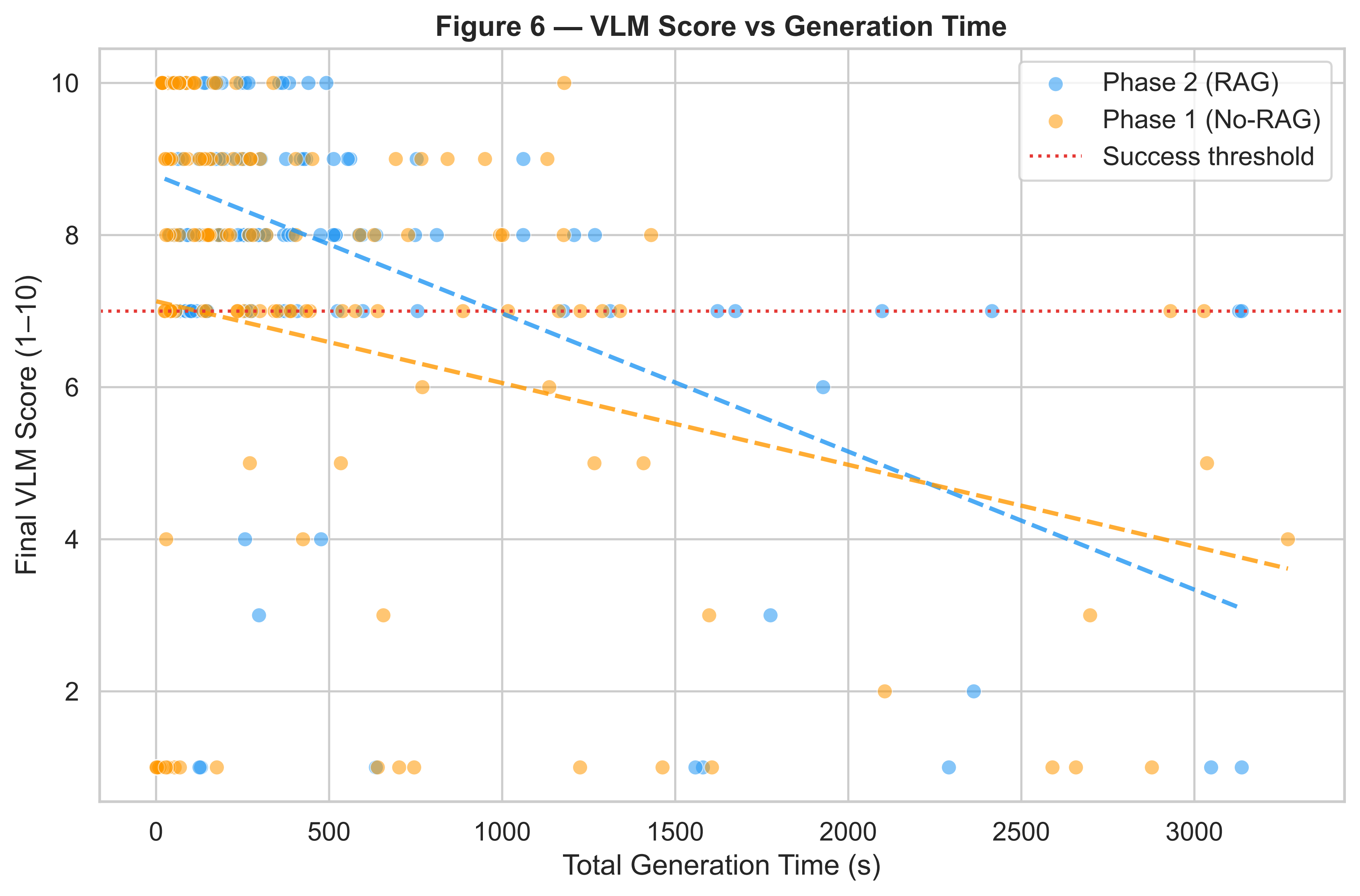


Figure 6 — VLM Score vs Generation Time
Phase 2 (RAG)
Phase 1 (No-RAG)
Success threshold
Final VLM Score (1–10)
Total Generation Time (s)
10
8
6
4
2
0
500
1000
1500
2000
2500
3000

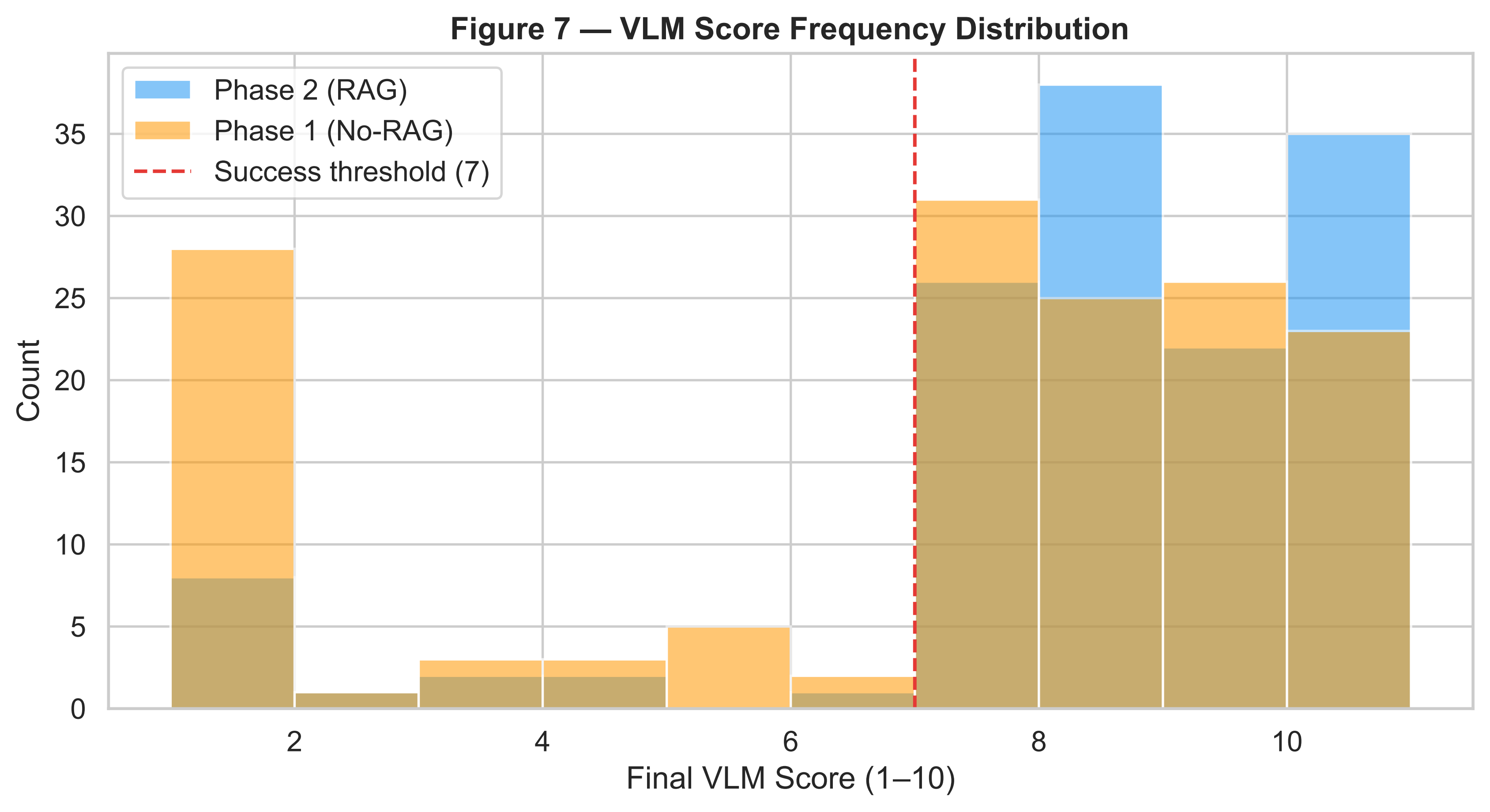


Figure 7 — VLM Score Frequency Distribution
Phase 2 (RAG)
Phase 1 (No-RAG)
Success threshold (7)
Count
0
5
10
15
20
25
30
35
2
4
6
8
10
Final VLM Score (1–10)

Figure 8 — Impact of Knowledge-Base Coverage (RAG Only)

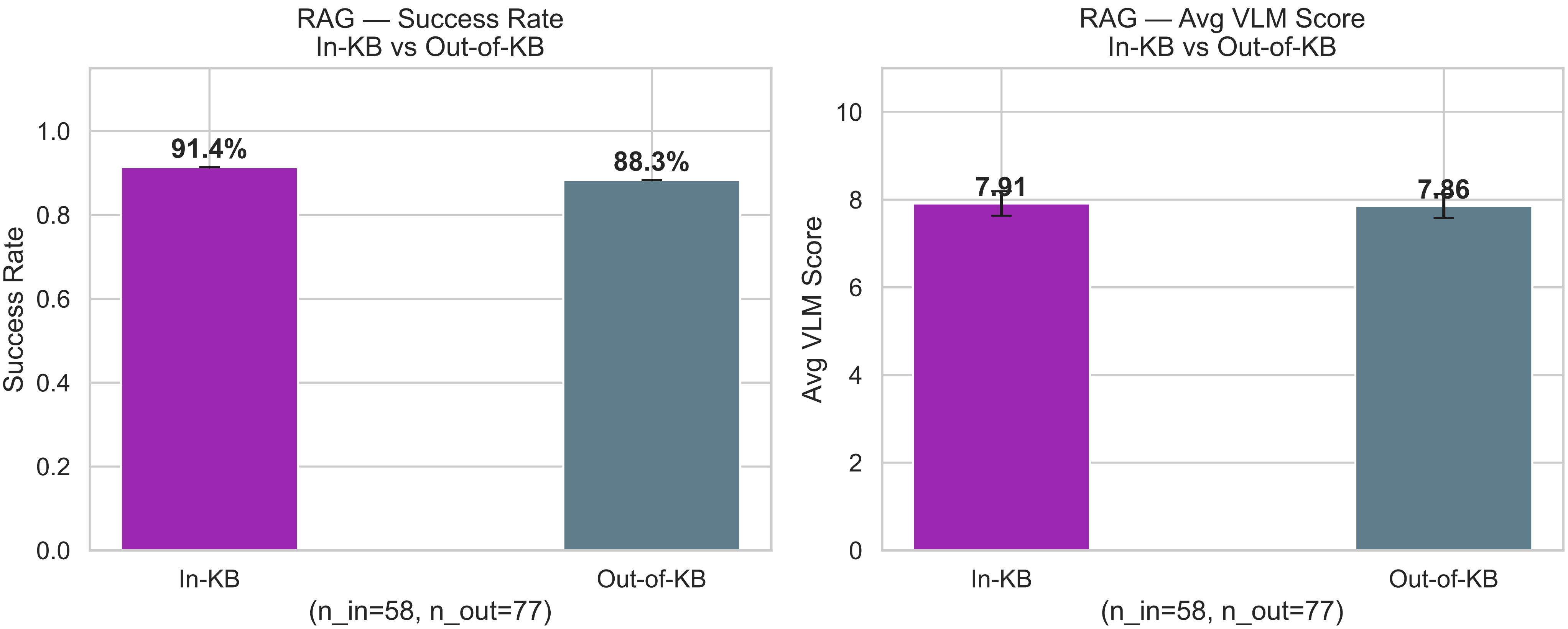
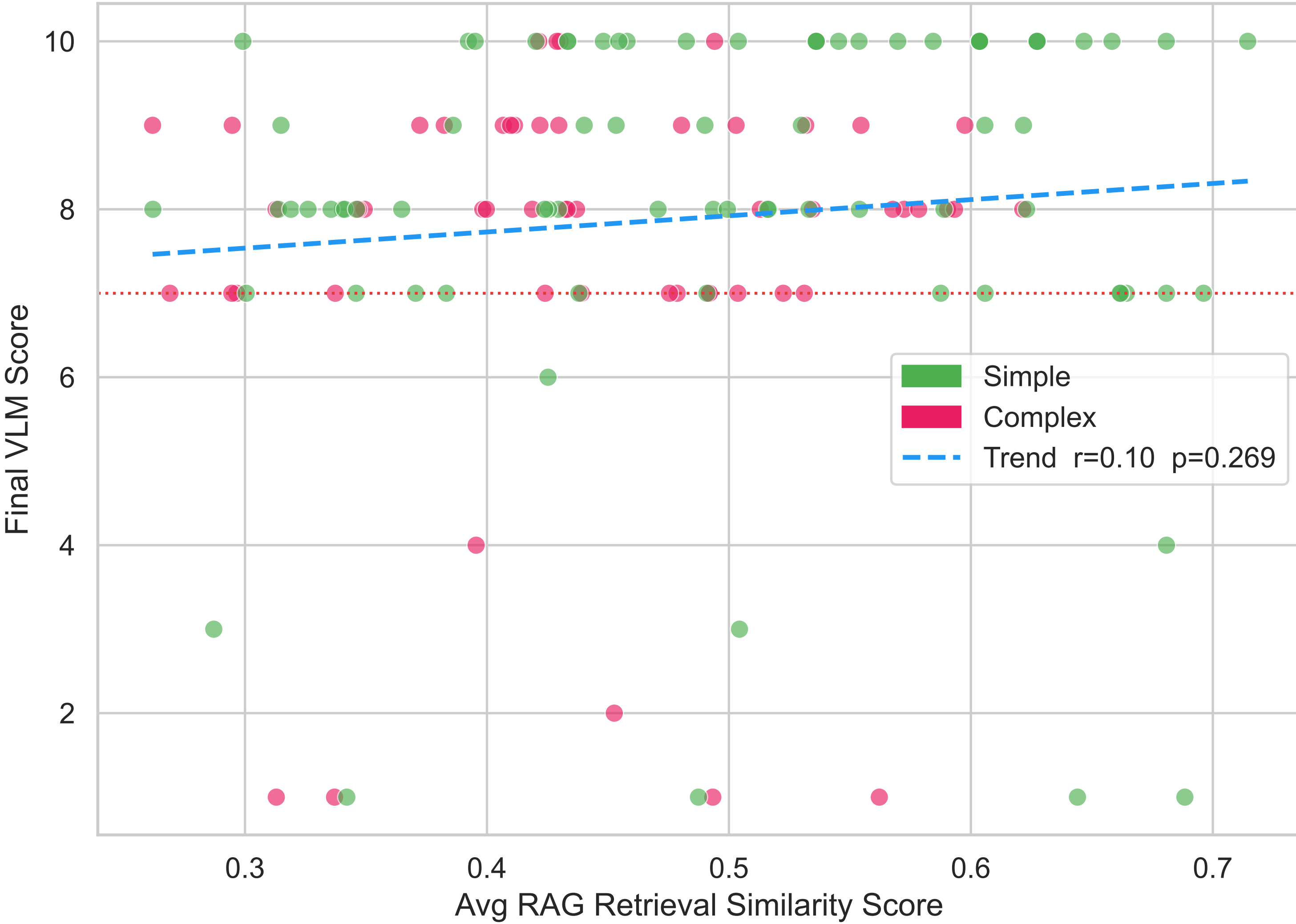


Figure 9 — Retrieval Similarity vs Generation Quality (RAG)



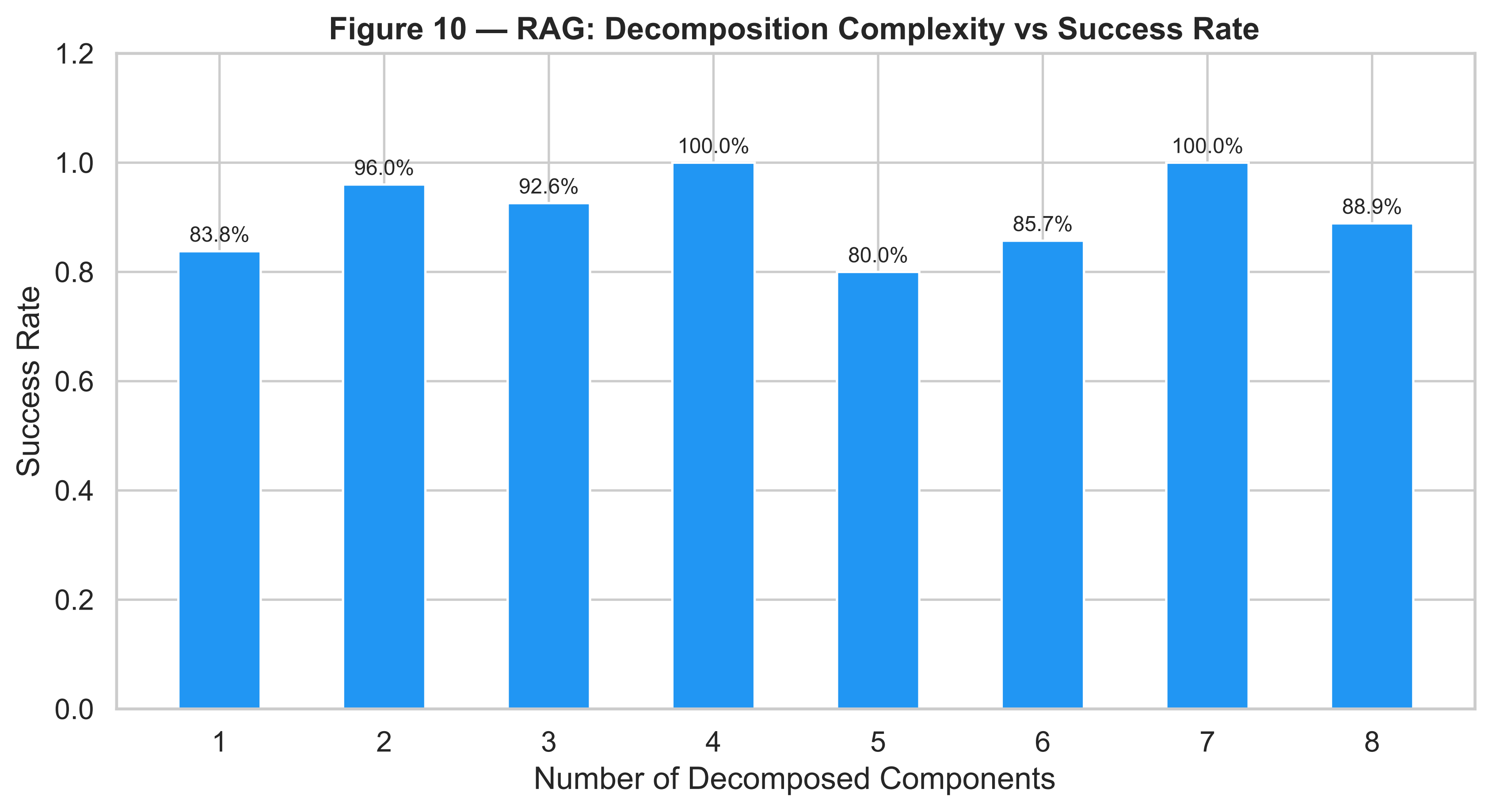


Figure 10 — RAG: Decomposition Complexity vs Success Rate
Success Rate
1.2
1.0
0.8
0.6
0.4
0.2
0.0
83.8%
96.0%
92.6%
100.0%
80.0%
85.7%
100.0%
88.9%
1
2
3
4
5
6
7
8
Number of Decomposed Components

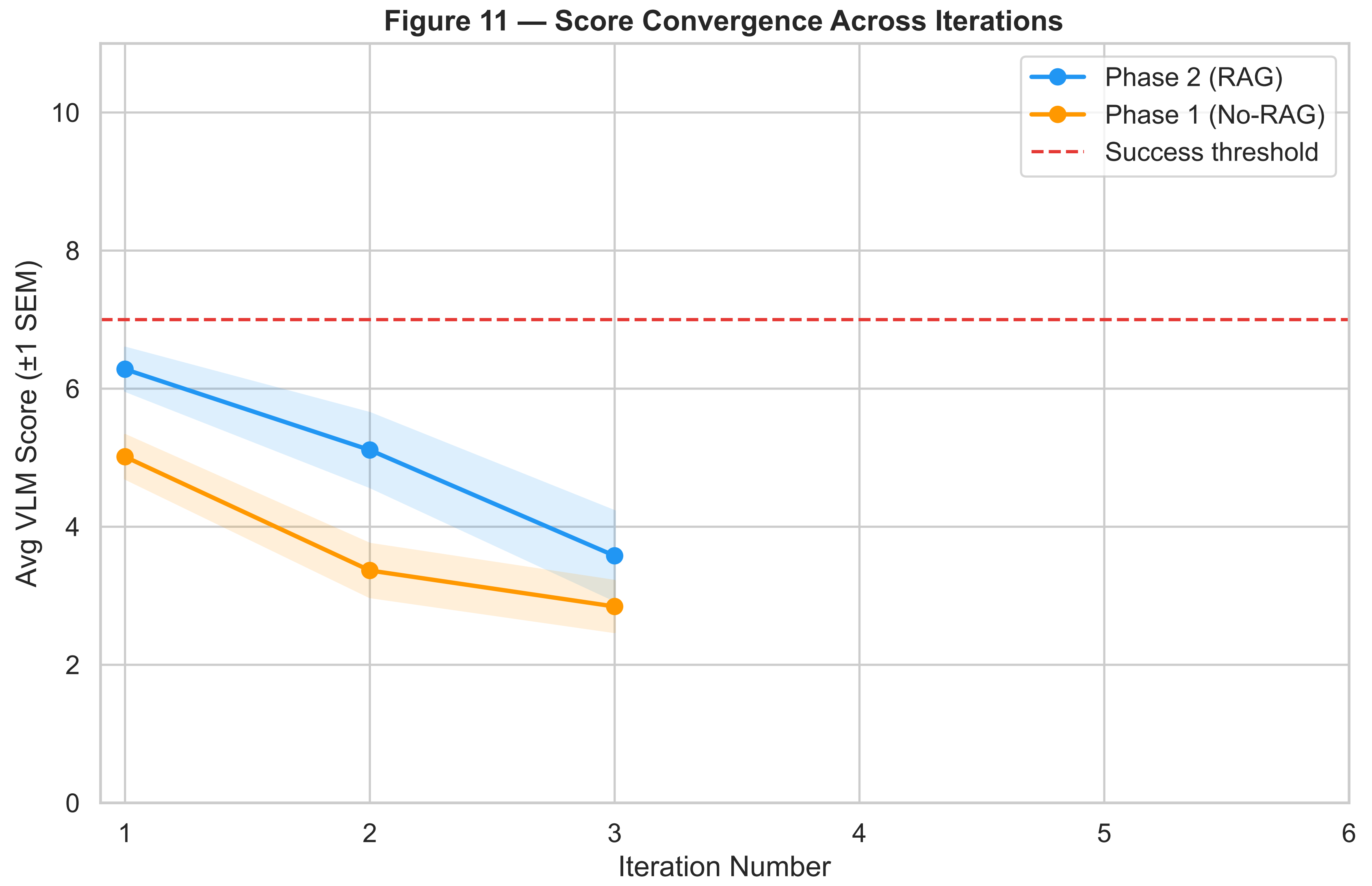


Figure 11 — Score Convergence Across Iterations
Phase 2 (RAG)
Phase 1 (No-RAG)
Success threshold
Avg VLM Score (±1 SEM)
Iteration Number
0
2
4
6
8
10
1
2
3
4
5
6

Figure 12 — Success Rate by Code LLM × Eval Provider

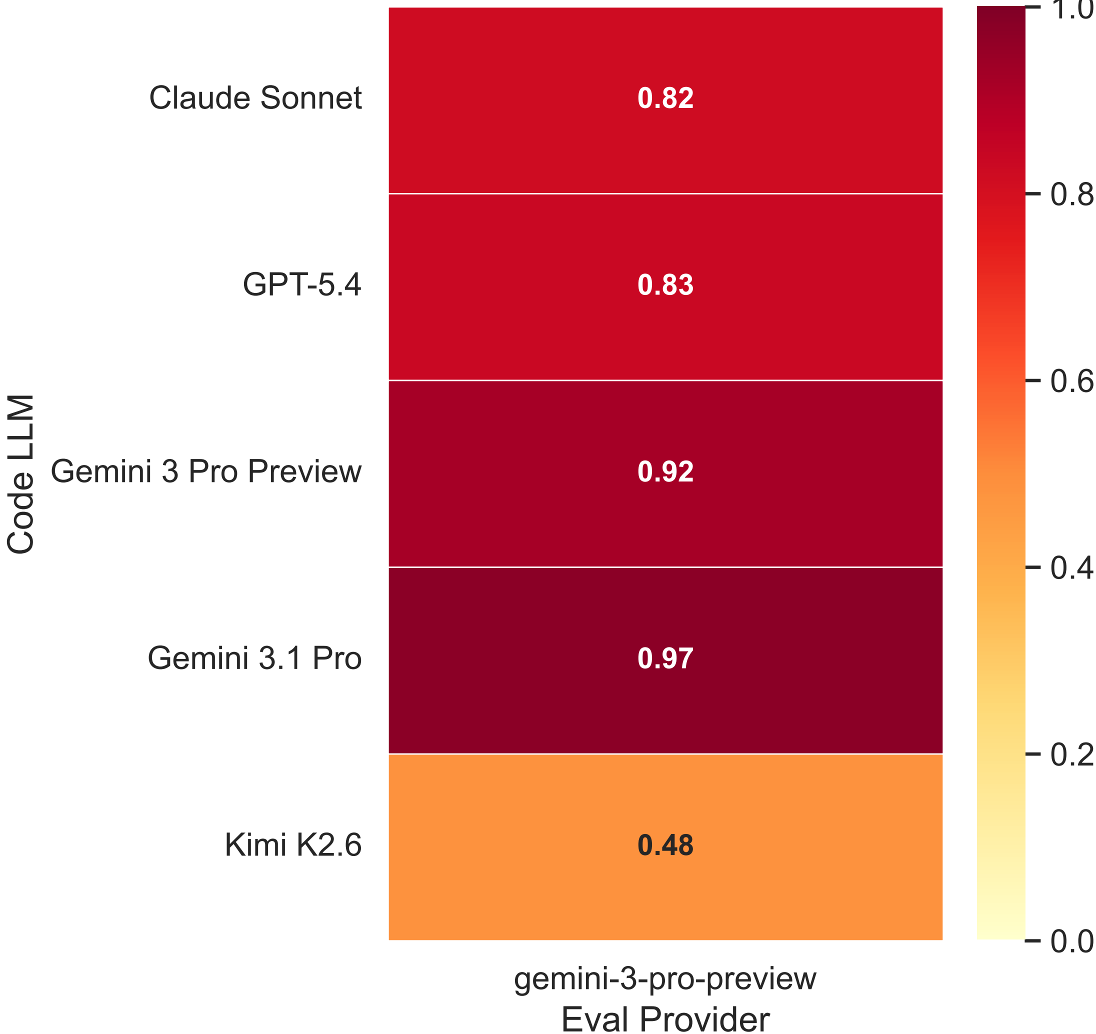


Figure 13 — VLM Score vs Human Evaluation

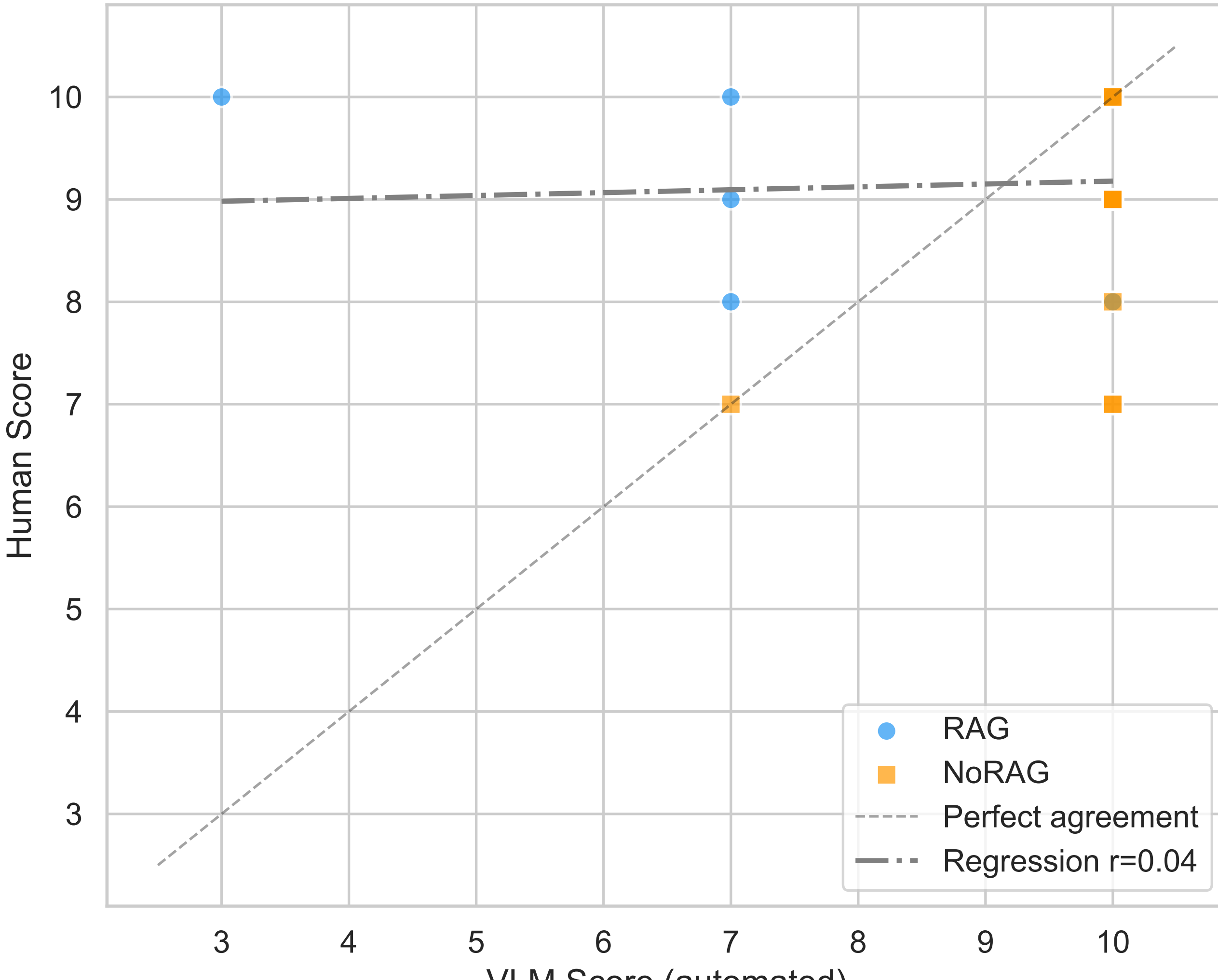
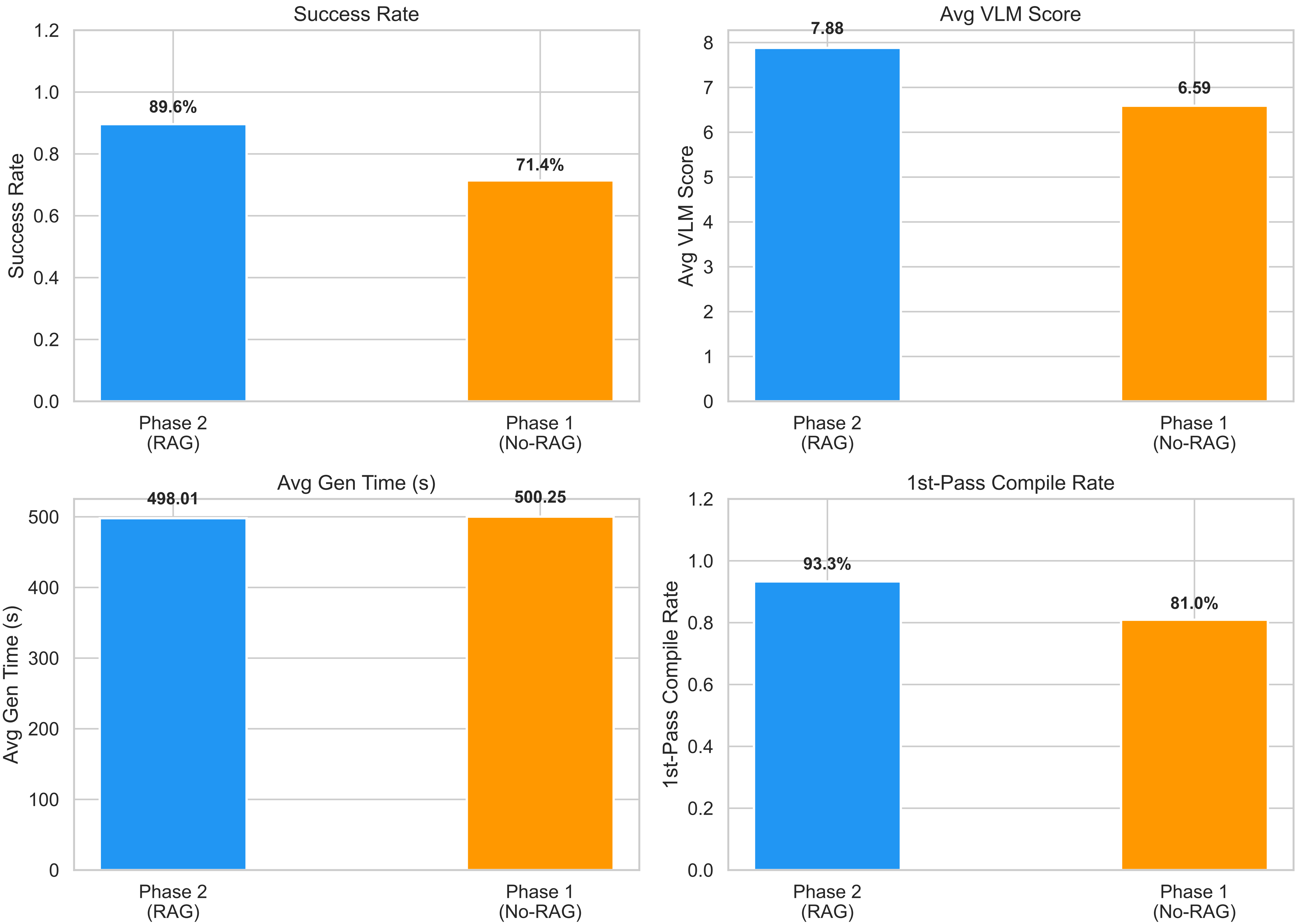


Figure 14 — Phase 1 vs Phase 2 Key Metrics Summary



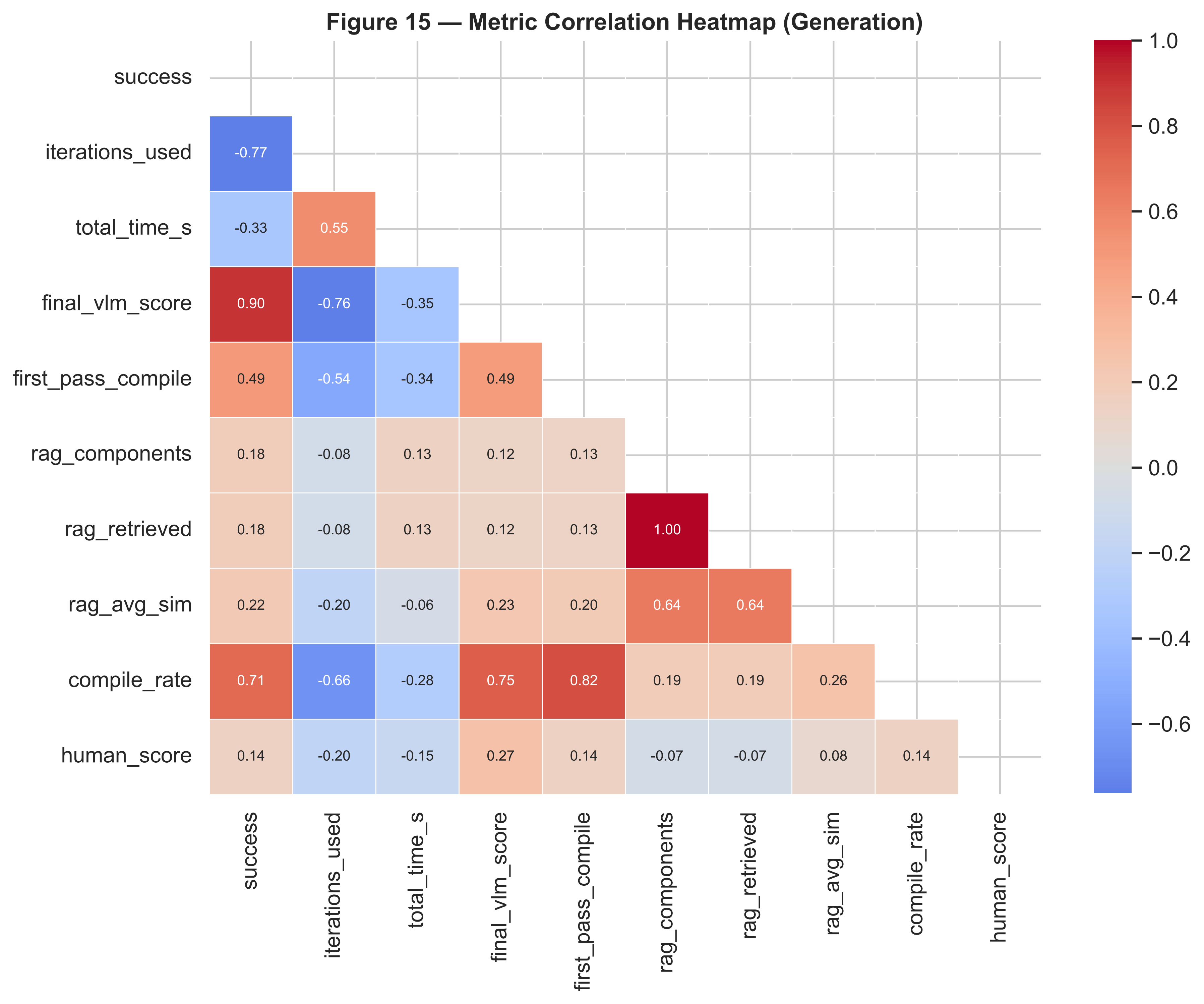
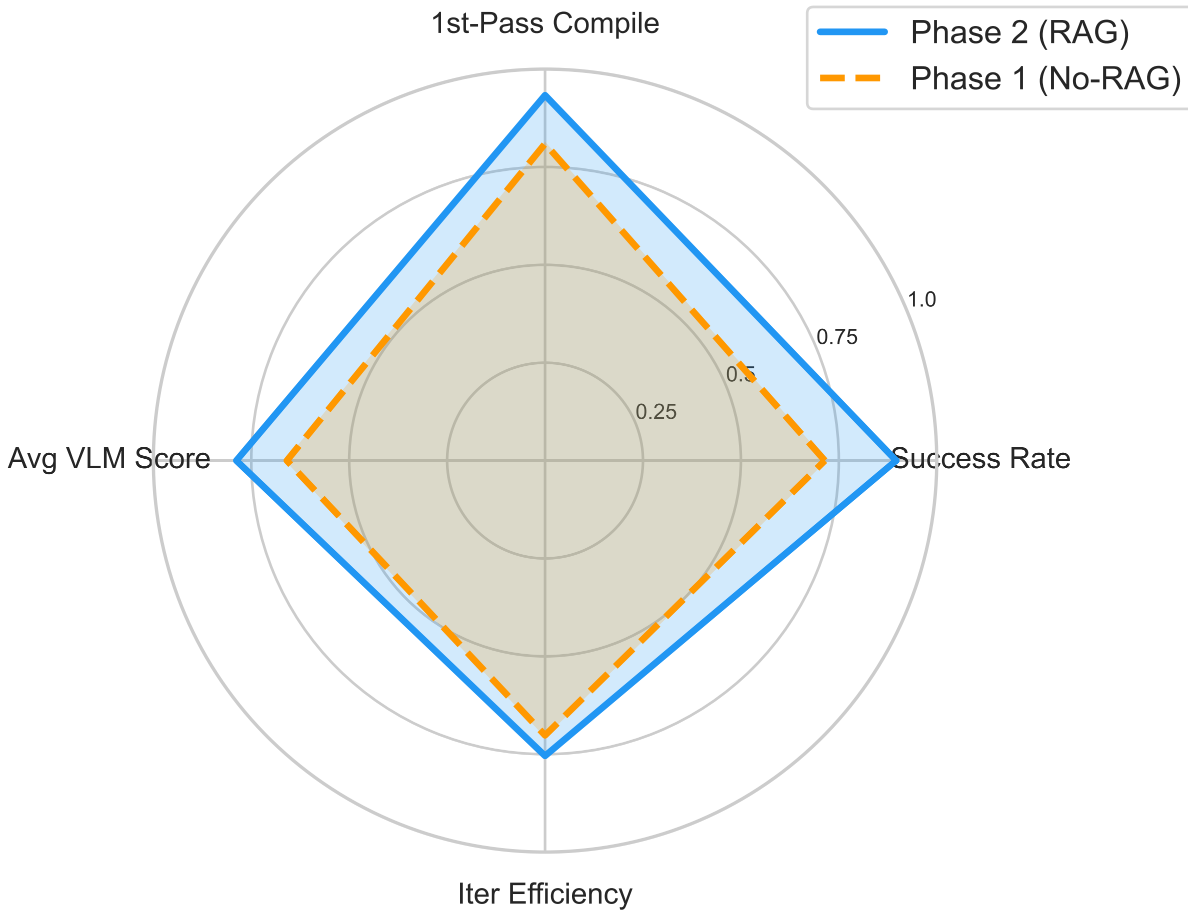


Figure 15 — Metric Correlation Heatmap (Generation)
success
iterations_used
total_time_s
final_vlm_score
first_pass_compile
rag_components
rag_retrieved
rag_avg_sim
compile_rate
human_score
-0.77
-0.33 0.55
0.90 -0.76 -0.35
0.49 -0.54 -0.34 0.49
0.18 -0.08 0.13 0.12 0.13
0.18 -0.08 0.13 0.12 0.13 1.00
0.22 -0.20 -0.06 0.23 0.20 0.64 0.64
0.71 -0.66 -0.28 0.75 0.82 0.19 0.19 0.26
0.14 -0.20 -0.15 0.27 0.14 -0.07 -0.07 0.08 0.14
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6

Figure 16 — Radar: Phase 1 vs Phase 2 Normalised Metrics



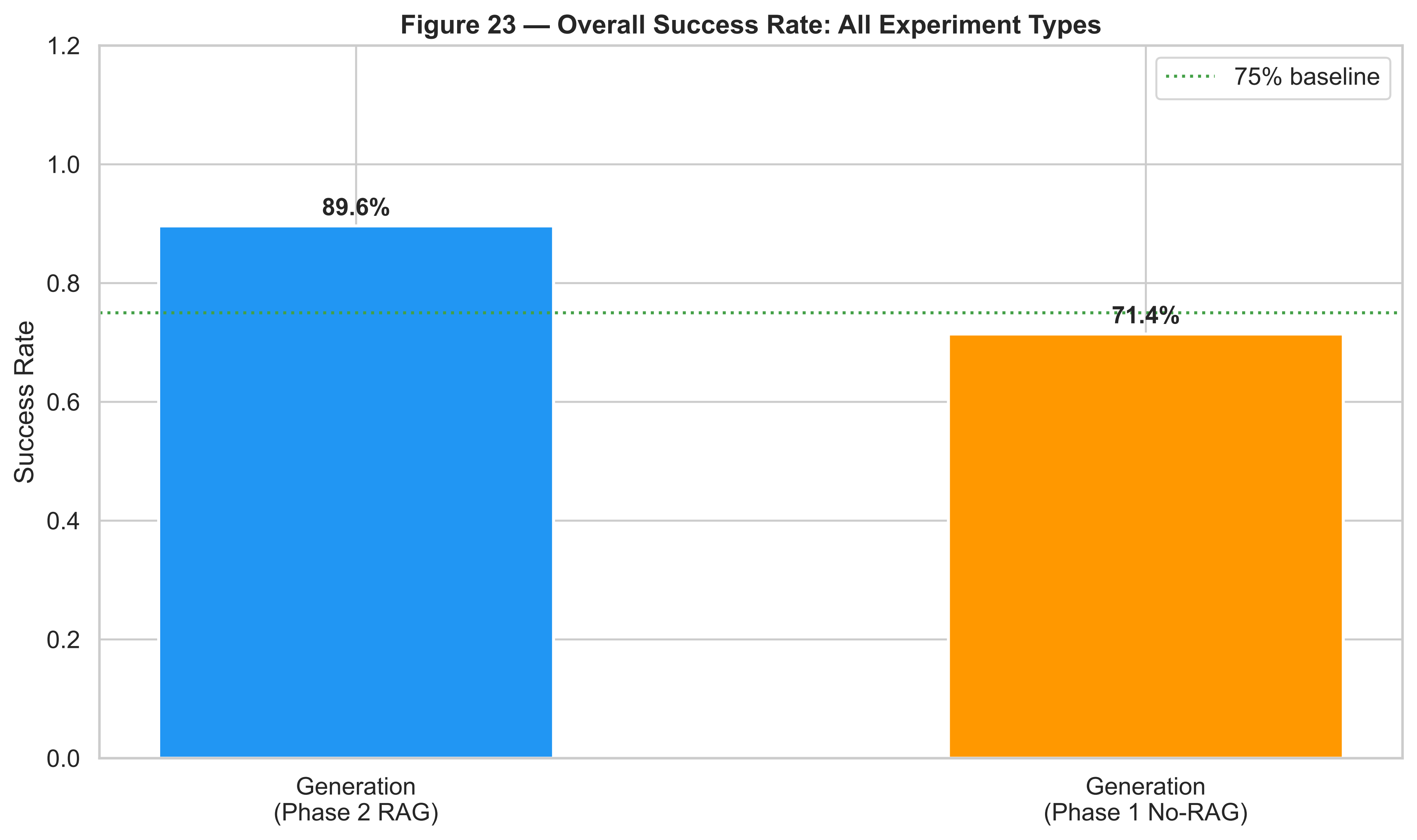


Figure 23 — Overall Success Rate: All Experiment Types
75% baseline
89.6%
71.4%
Success Rate
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Generation
(Phase 2 RAG)
Generation
(Phase 1 No-RAG)

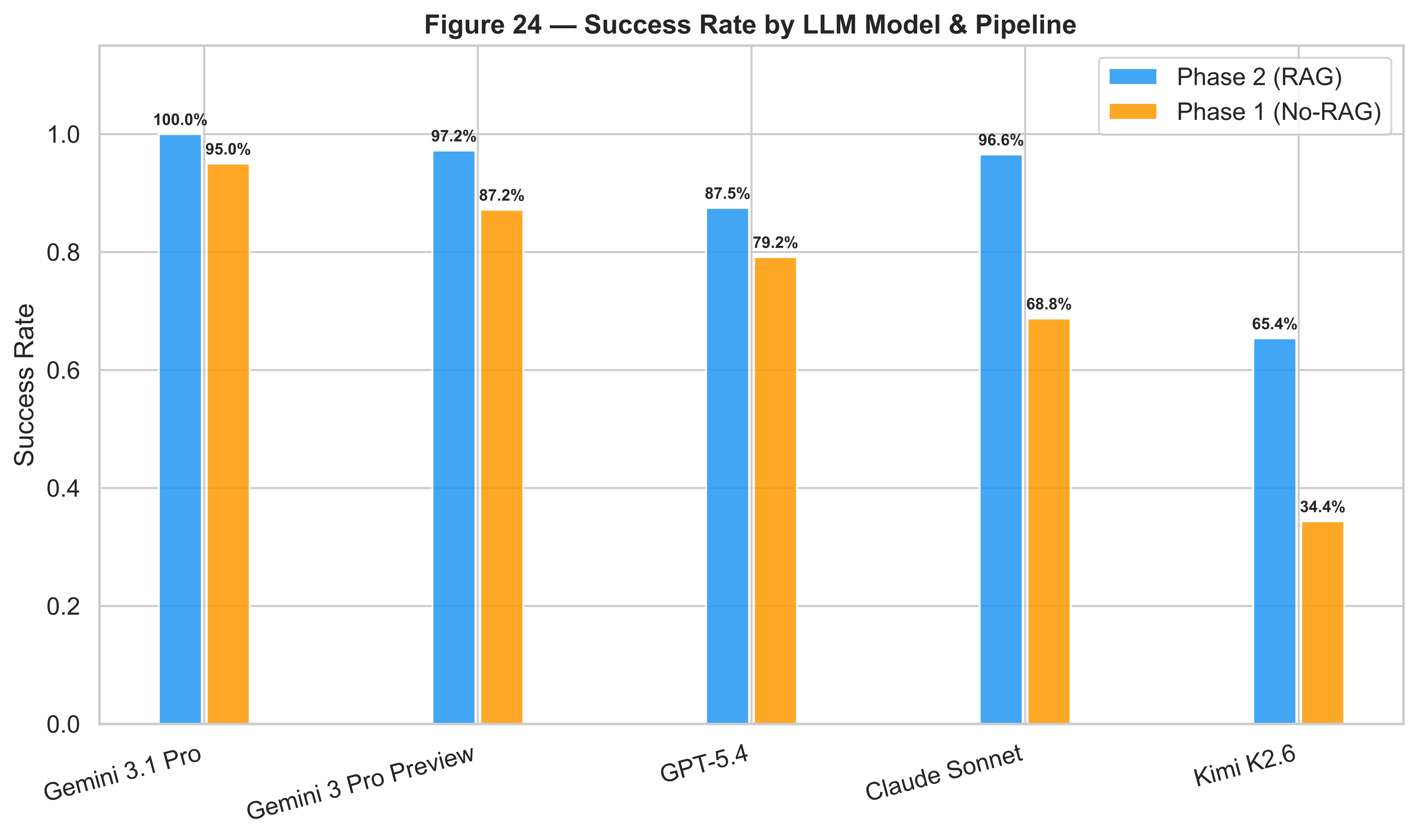


Figure 24 — Success Rate by LLM Model & Pipeline
Phase 2 (RAG)
Phase 1 (No-RAG)
Success Rate
1.0
0.8
0.6
0.4
0.2
0.0
100.0%
95.0%
97.2%
87.2%
87.5%
79.2%
96.6%
68.8%
65.4%
34.4%
Gemini 3.1 Pro
Gemini 3 Pro Preview
GPT-5.4
Claude Sonnet
Kimi K2.6

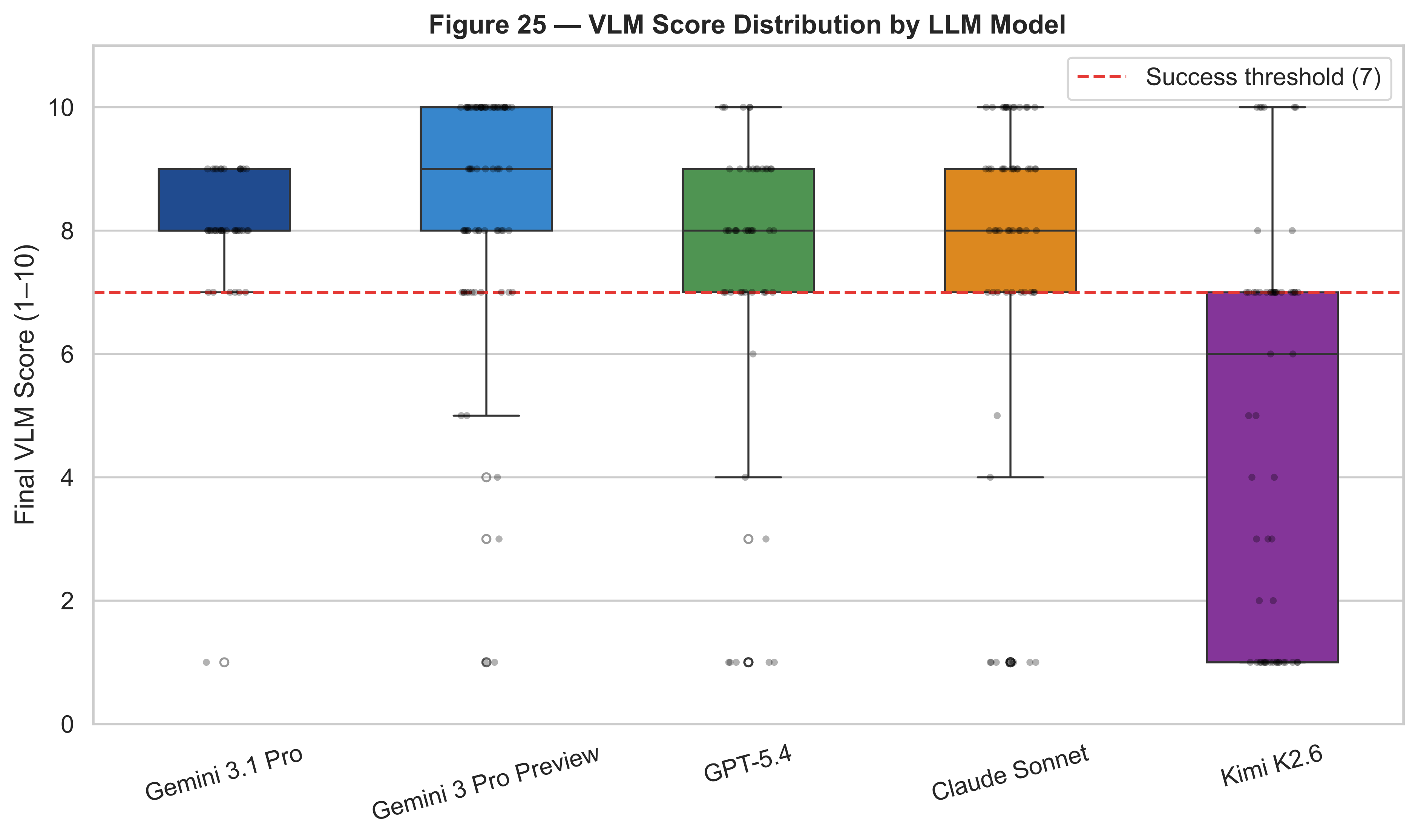


Figure 25 — VLM Score Distribution by LLM Model
Success threshold (7)
Final VLM Score (1–10)
0
2
4
6
8
10
Gemini 3.1 Pro
Gemini 3 Pro Preview
GPT-5.4
Claude Sonnet
Kimi K2.6

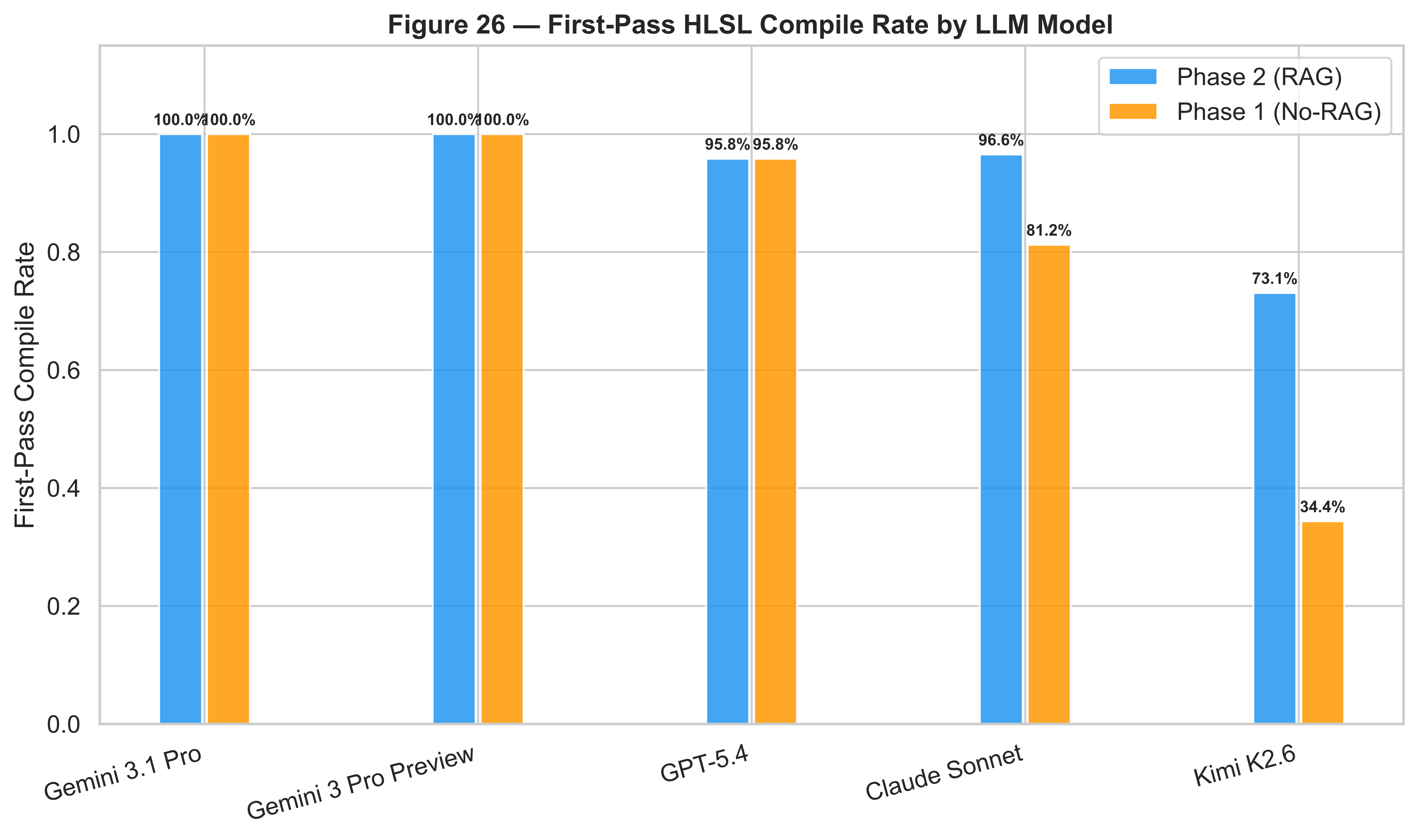


Figure 26 — First-Pass HLSL Compile Rate by LLM Model
Phase 2 (RAG)
Phase 1 (No-RAG)
First-Pass Compile Rate
0.0
0.2
0.4
0.6
0.8
1.0
100.0%
100.0%
100.0%
100.0%
95.8%
95.8%
96.6%
81.2%
73.1%
34.4%
Gemini 3.1 Pro
Gemini 3 Pro Preview
GPT-5.4
Claude Sonnet
Kimi K2.6

Figure 27 — Average Iterations to Success by LLM Model

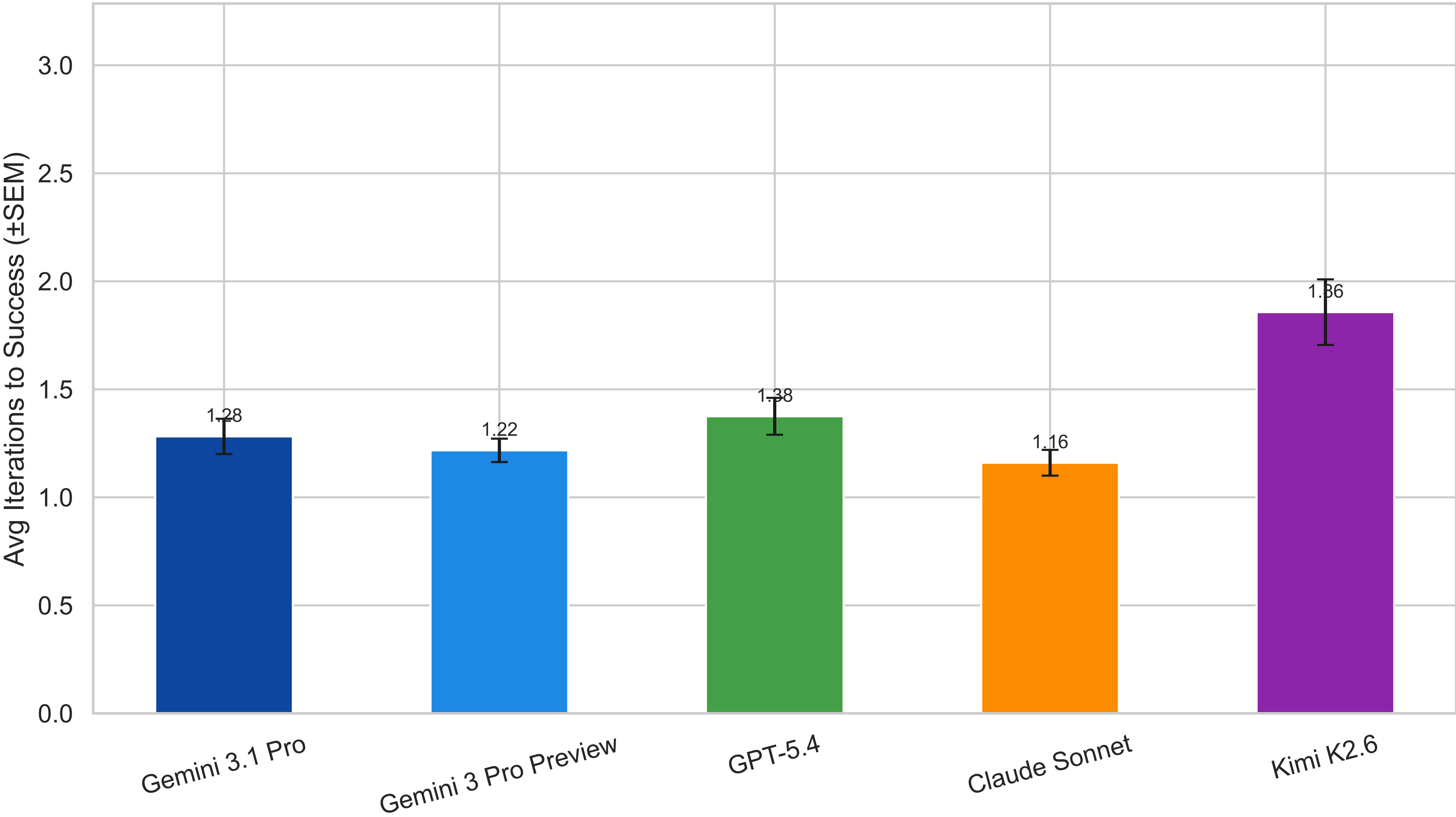


Figure 28 — Generation Time by LLM Model

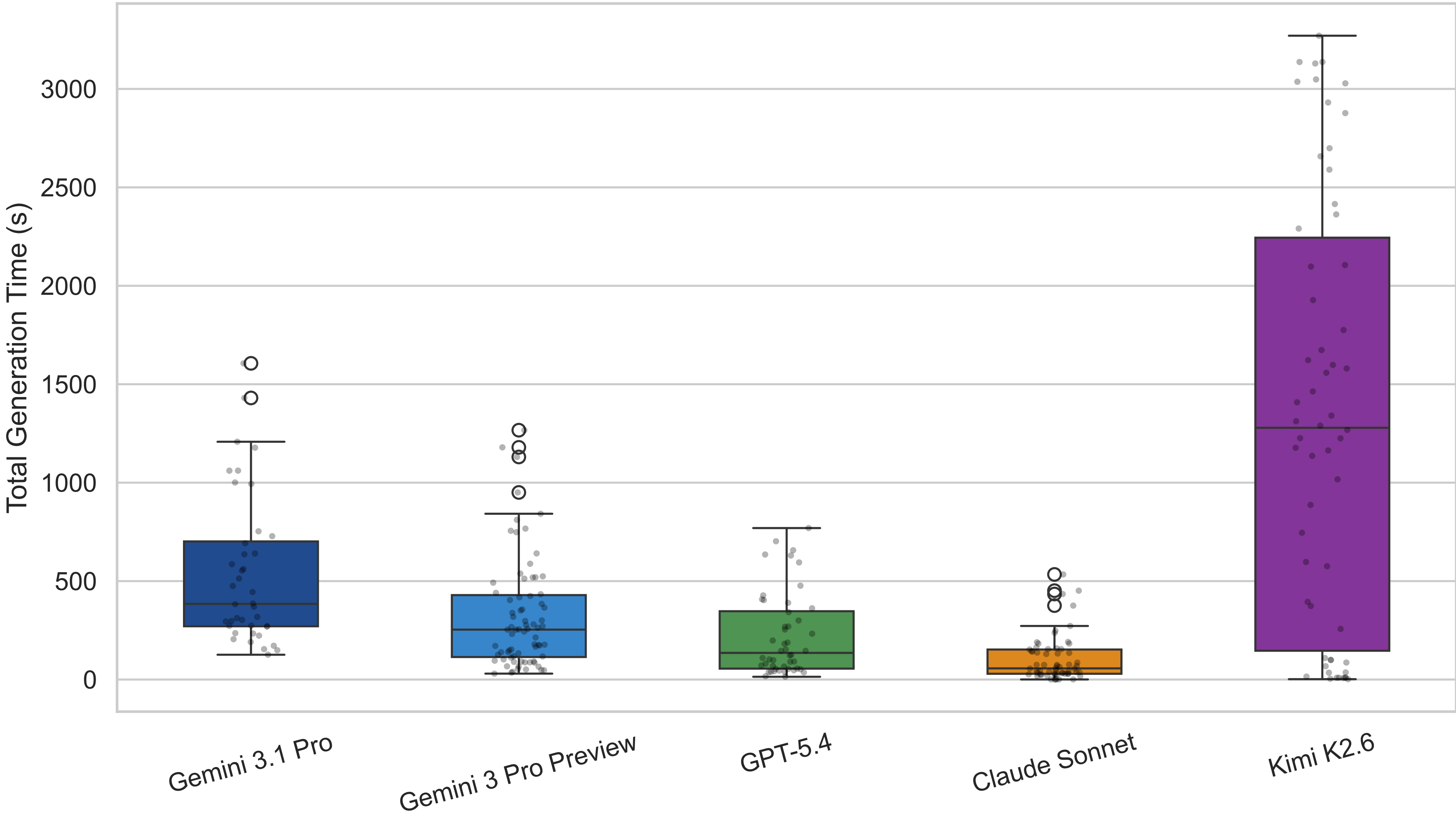
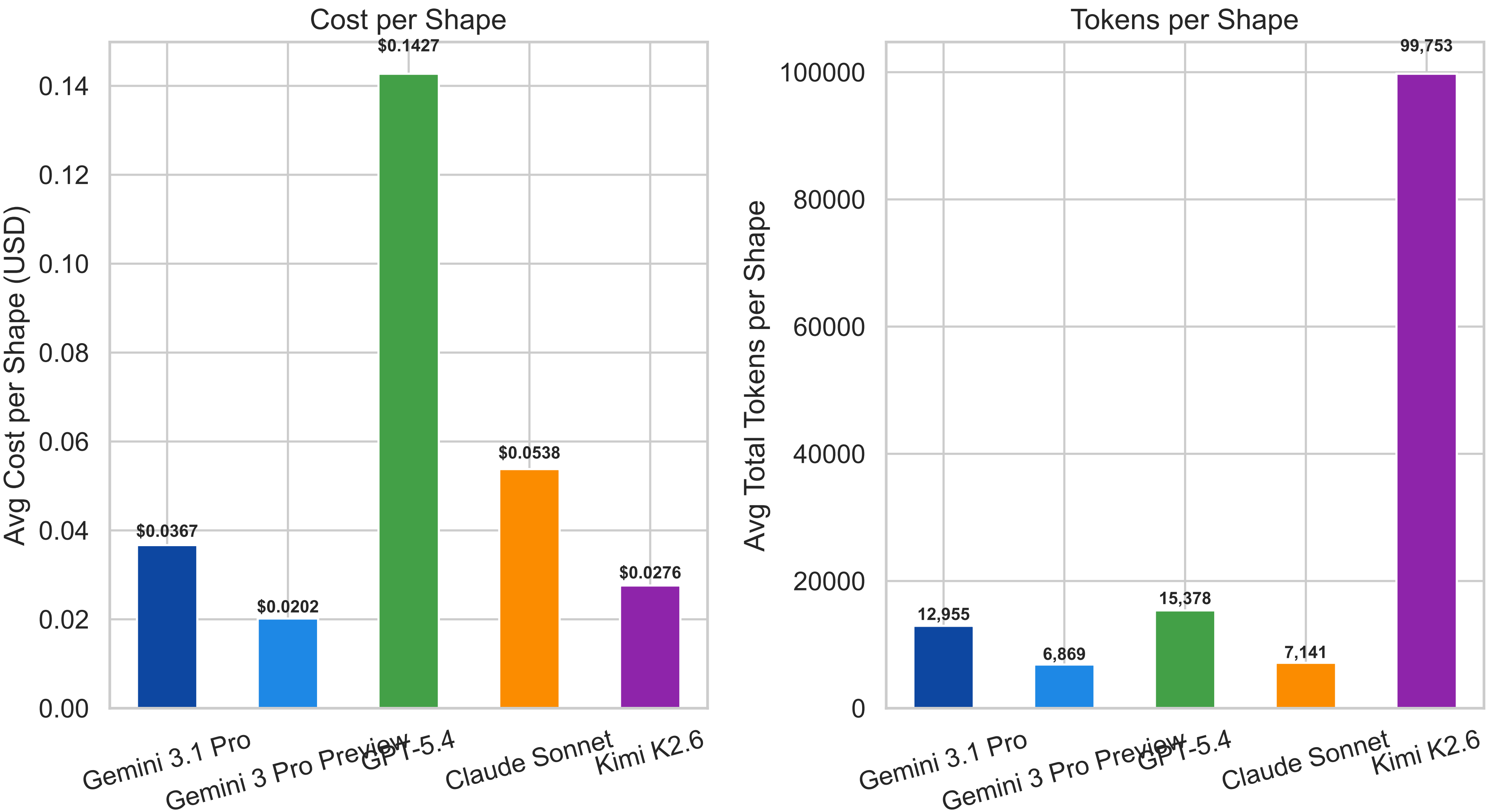


Figure 29 — Cost & Token Usage by LLM Model



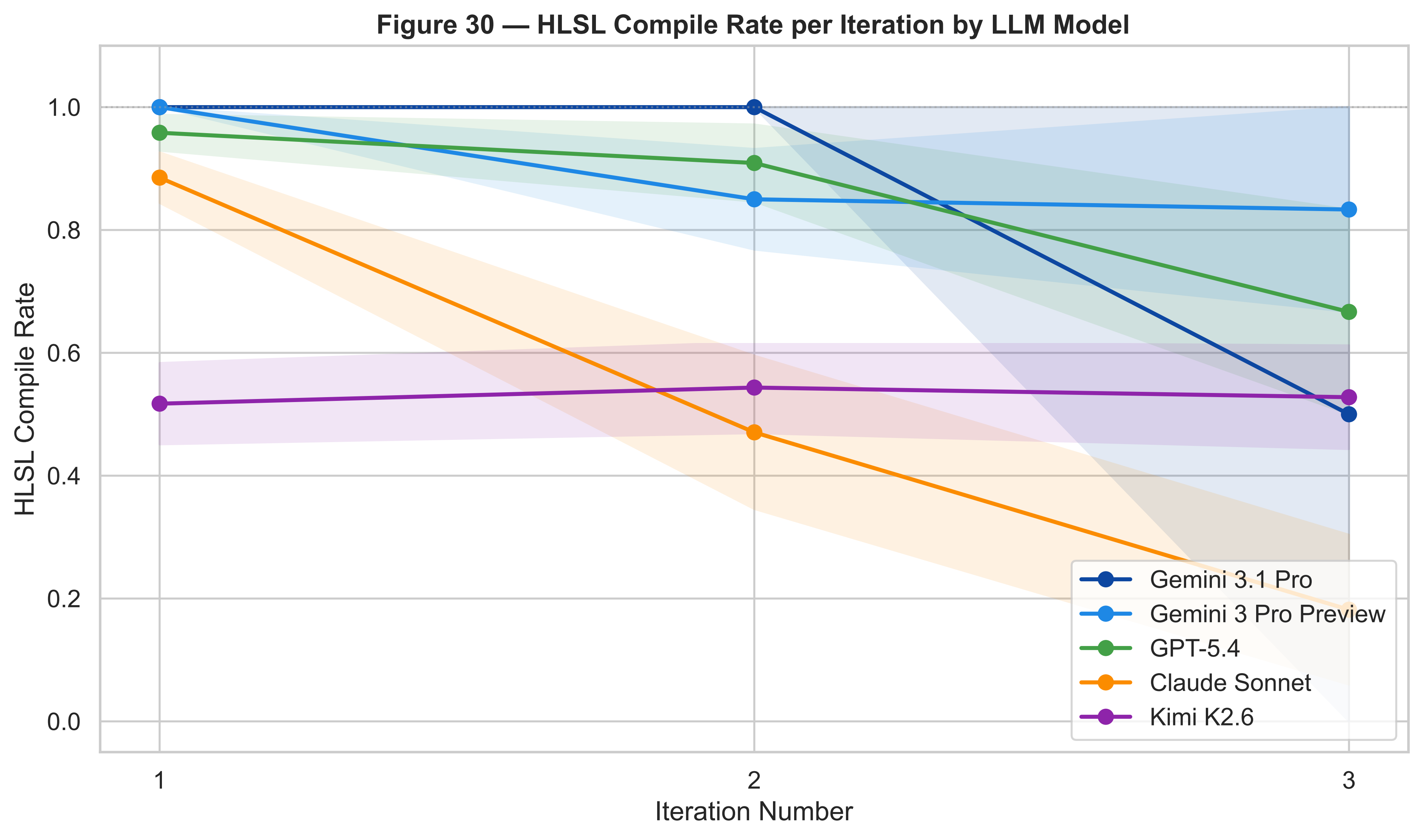


Figure 30 — HLSL Compile Rate per Iteration by LLM Model
HLSL Compile Rate
Iteration Number
0.0
0.2
0.4
0.6
0.8
1.0
1
2
3
Gemini 3.1 Pro
Gemini 3 Pro Preview
GPT-5.4
Claude Sonnet
Kimi K2.6

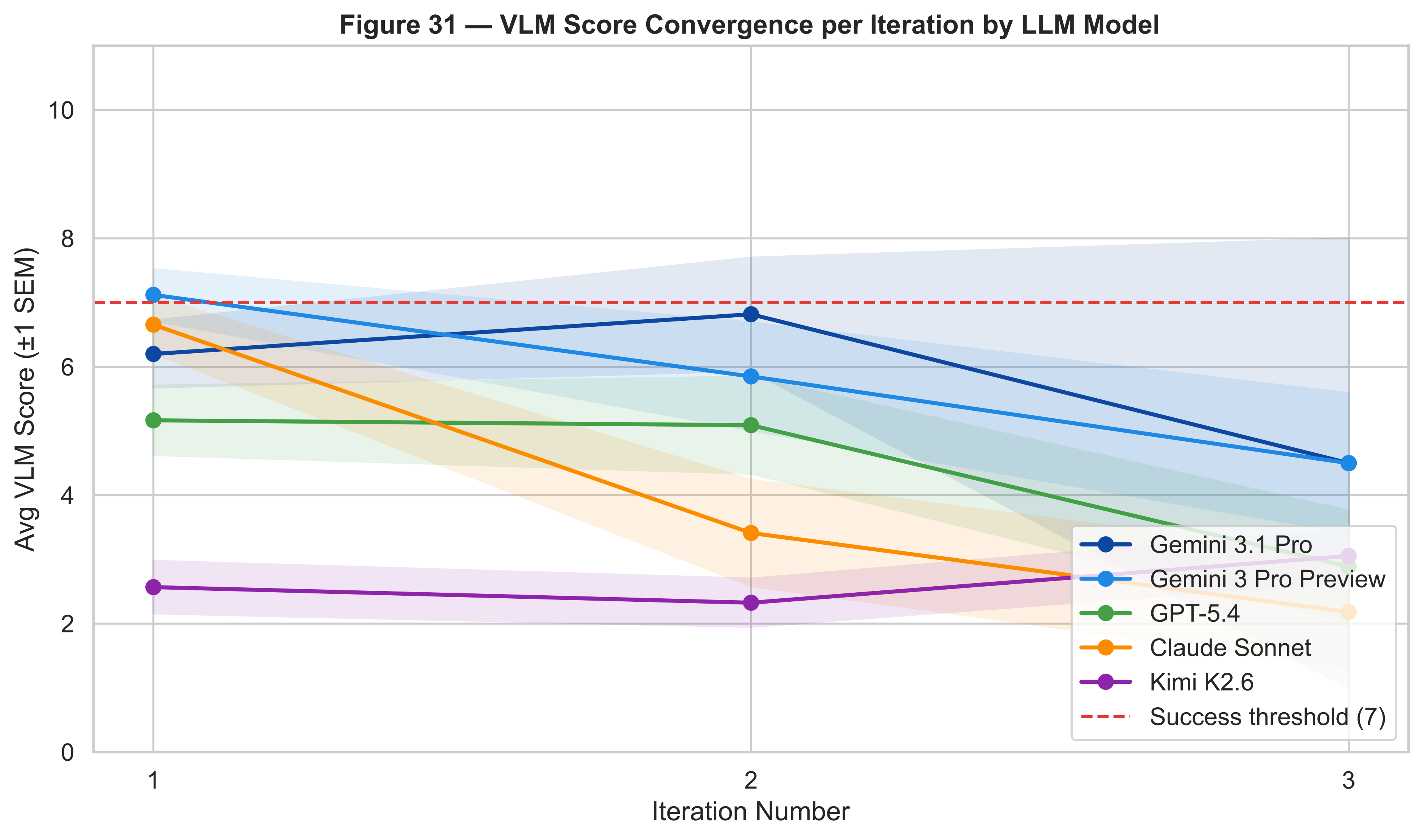


Figure 31 — VLM Score Convergence per Iteration by LLM Model
Avg VLM Score (±1 SEM)
Iteration Number
0
2
4
6
8
10
1
2
3
Gemini 3.1 Pro
Gemini 3 Pro Preview
GPT-5.4
Claude Sonnet
Kimi K2.6
Success threshold (7)

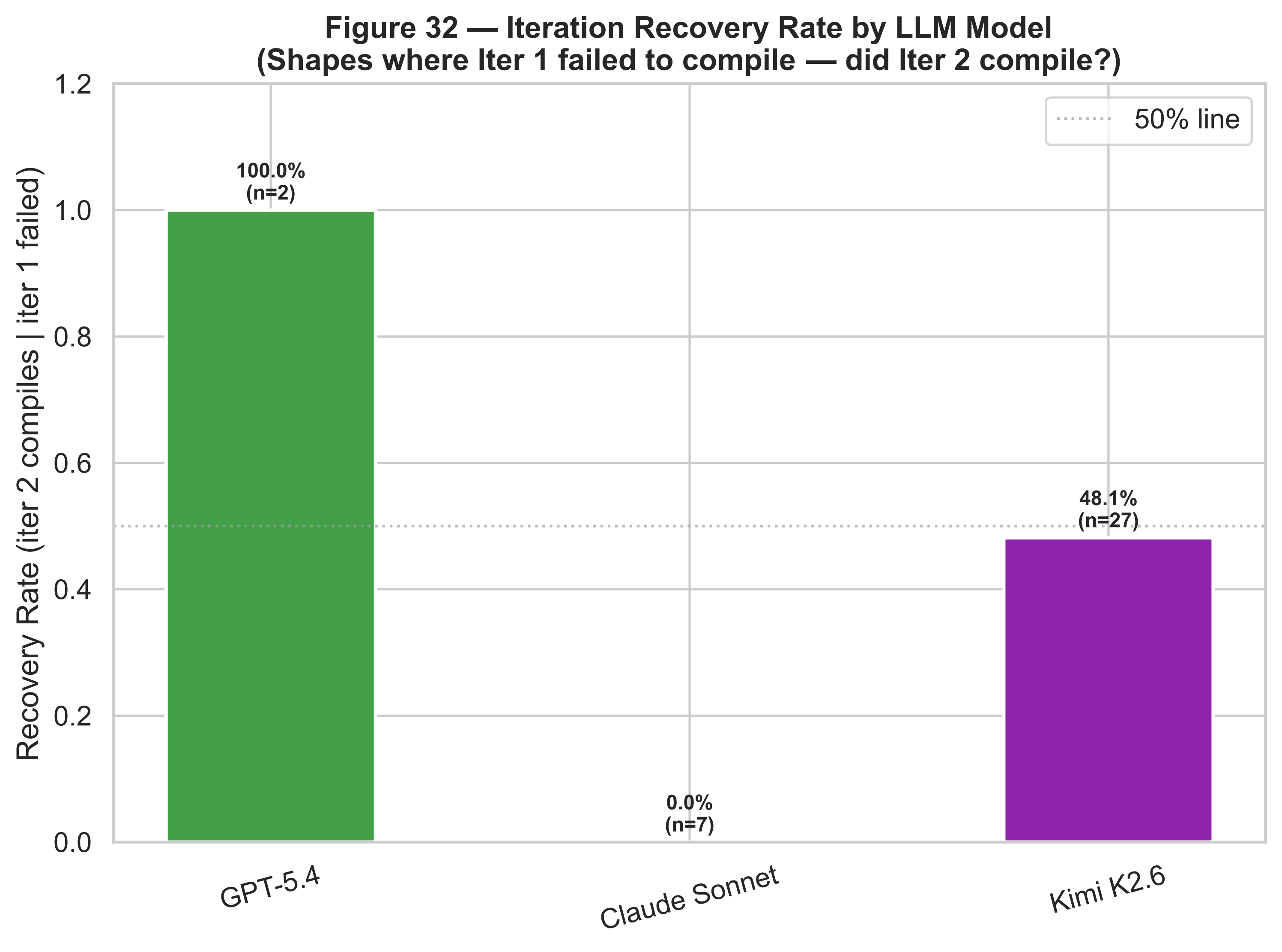


Figure 32 — Iteration Recovery Rate by LLM Model
(Shapes where Iter 1 failed to compile — did Iter 2 compile?)
Recovery Rate (iter 2 compiles | iter 1 failed)
50% line
100.0%
(n=2)
0.0%
(n=7)
48.1%
(n=27)
1.2
1.0
0.8
0.6
0.4
0.2
0.0
GPT-5.4
Claude Sonnet
Kimi K2.6